MILITARY SPECIFICATION

SPRING, HELICAL TORSION

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of helical torsion spring.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-H-1581	Hardware (Fasteners and Related Items), Packaging of

STANDARDS

MILITARY

MIL-STD-29	Springs, Mechanical, Drawing Requirements For
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1420
5360

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PUBLICATIONS

WR 43A	Preparation of Quality Assurance Provisions

DRAWING

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 10001)

2841637	Spring, Helical Torsion

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article. When specified in the contract (see 6.2), the contractor shall furnish sample springs for first article inspection and approval (see 4.5).

3.2 Design. The spring shall be a left hand helix open coil helical torsion spring. The design shall be in accordance with Dwg. 2841637.

NOTE: The 4½ REF Number of coils-free position and the 180° REF tang angle specified on Dwg. 2841637 are not design parameters. The number of turns and the resulting tang angle are calculated using spring formulae and the torque requirements specified.

3.3 Spring formulae. For design information and formulae, the contractor shall refer to MIL-STD-29.

3.4 Material. The spring shall be formed from steel wire as specified on Dwg. 2841637.

NOTE: The forming operation shall allow for relaxation of the springs during the stress relief operation.

3.5 Stress relief. The spring shall be stress relieved at 260 to 290°C (500 to 550°F) for 30 minutes minimum, after forming, but prior to plating.

3.6 Protective coating. The spring shall be cadmium plated in accordance with Type II, Class 2 of QQ-P-416.

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3.7 Hydrogen embrittlement relief (M101). The cadmium plated spring shall be baked after plating at 175 to 205°C (350 to 400°F) for a minimum of 24 hours for hydrogen embrittlement relief. Baking shall be initiated within 8 hours of plating. The spring must not be flexed prior to baking.

3.8 Mechanical requirements (M102). The spring shall operate over a .750 to .753 inch diameter shaft and develop a torque of 8.5 to 10.0 in, lb when one tang is held stationary and the other is rotated to the point representing the initial position on Dwg. 2841637 when tested in accordance with 4.7

3.9 Workmanship. The spring shall be uniform in quality and shall be free from cracks, irregularities, scale, surface imperfections, burrs or slivers which could adversely affect safety or performance. Sharp edges shall be rounded equivalent to .015 R maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Humidity: Room ambient to 95 percent relative, maximum

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5)
b. Quality conformance inspection (see 4.6)

4.4 Sampling.

4.4.1 First article sample. When specified in the contract (see 6.2), a first article sample of ten springs, manufactured in accordance with 3.2, shall be subjected to the first article inspection detailed in 4.5 at an activity designated by the procuring activity (see 6.2). Acceptance of the first article sample shall be based on no defects in the sample. Further production of springs by the contractor prior to approval of the first article sample shall be at the contractor's risk.

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4.4.2 Lot. A lot shall consist of all springs, manufactured under essentially the same conditions, from the same wire stock, by one manufacturer, at one plant, and to be offered for delivery at one time.

4.4.3 *Sampling for quality conformance inspection.*

4.4.3.1 Sampling for test. Sampling for torque tests shall be in accordance with inspection level III of MIL-STD-105.

4.4.3.2 Sampling for examination. Sampling for examination shall be in accordance with inspection level II of MIL-STD-105.

4.5 First article inspection. The first article inspection shall consist of the torque test of 4.7 and the visual and dimensional examination of 4.8 performed on the sample springs specified in 4.4.1. Failure of any first article spring shall be grounds for rejection of the first article sample.

4.6 Quality conformance inspection. The quality conformance inspection shall consist of the torque test of 4.7 and the examinations of 4.8 and 4.9. The acceptable quality levels (AQL's) shall be 2.5 percent defective for tests and 6.5 defects per one hundred springs for examinations. Individual springs failing the torque test shall not be delivered on contract.

4.6.1 Classification of characteristics. The characteristics verified by the tests and examination herein are classified as critical, major, or minor in accordance with WR-43A. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test of examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.7 Torque tests. The torque test shall be performed in accordance with the following using a torque tester in accordance with FIGURE 1 as approved by the procuring activity prior to testing (see 6.2):

a. Mount the torque tester vertically in a vice so that the plane of the wheel is vertical and the weights hang unobstructed.

b. Place the sample spring on the .750 to .753 inch diameter test shaft, insert the horizontal tang into the guide hole in the end plate and tighten the end plate screws taking care that the end plate does not contact the shaft.

of the first article sample shall be based on no defects in the sample. Further production of springs by the contractor prior to approval of the first article sample shall be at the contractor's risk.

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d. Attach a 3.40 ± .05 lb weight to the cable, tap the base plate with the hand several times to ensure an accurate reading, and record the scale reading.

e. Remove the 3.4 lb weight and substitute a 4.00 ± .05 lb weight, tap the plate several times, and record the scale reading.

f. Acceptable readings at 3.4 lb and 4.0 lb are 27 to 43° and 30 to 52°, respectively. Reject any spring that has a final reading outside the specified range at 3.4 lbs or 4.0 lbs.

4.8 Visual and dimensional examination. Each sample spring selected in accordance with 4.4.3 shall be visually examined to verify conformance with 3.9 and dimensionally examined to verify conformance with Dwg. 2841637.

4.9 Packaging examination. Examination of packaging to verify conformance with Section 5 shall be in accordance with PPP-H-1581.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C in accordance with PPP-H-1581 as specified in the contract (see 6.2).

5.2 Packing. Packing shall be level A, B or C in accordance with PPP-H-1581 as specified in the contract (see 6.2).

5.3 Marking. In addition to any special markings required by the contract (see 6.2), unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Springs in accordance with this specification are intended for use in the Mk 276 Mods 0 and 1 igniter and arming mechanism assemblies (Dwgs. 2841587 and 2806546) and for the Mk 56 Mods 0, 1 and 2 rocket motor assemblies (Dwgs. 2841586, 2807085 and 3274407).

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Quantity required.
c. Place of delivery.
d. When a first article sample is required (see 3.1, 4.4.1 and 6.3).
e. Inspection conditions when other than as specified (see 4.2).
f. Assigned activity for first article inspection (see 4.4.1 and 6.3).

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g. Level of preservation and packaging required (see 5.1).
h. Level of packing required (see 5.2).
i. Any special markings required (see 5.3).

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD 1664) which will be documented in the Applicable ADL (Authorized Data List).

6.3 First article sample. When a first article sample is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulations (ASPR). The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article sample.

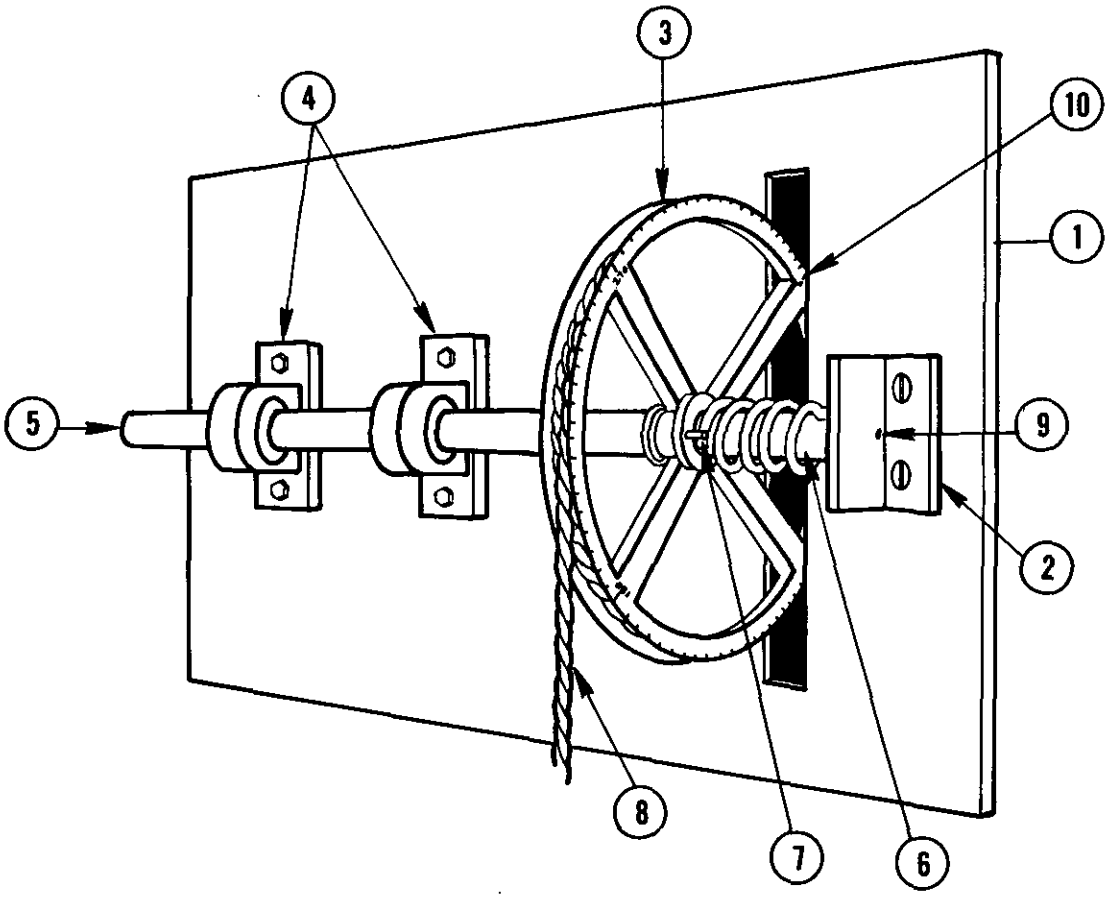
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HELICAL TORSION SPRING CALIBRATION DEVICE



1. BASE PLATE
2. END PLATE (MOVEABLE, METHOD OPTIONAL)
3. WHEEL, CALIBRATED 360° IN 1° INCREMENTS, 2.50 ± 0.5 RADIUS TO CENTERLINE OF CABLE, .68 TO .70 INCHES FROM END PLATE
4. PILLOW BLOCK (2), ANTIFRICTION BALL BEARING. ID COMPATIBLE WITH SHAFT OD
5. SHAFT, DIAMETER ≤.750 INCHES
6. TEST SLEEVE, OD .750 TO .753 INCHES; ID EQUAL TO SHAFT DIAMETER
7. CONTACT PIN, 0.35 MINIMUM LENGTH, .102 NOMINAL DIAMETER
8. CABLE, ATTACHED TO WHEEL TO ALLOW TORQUE LOADING
9. TANG GUIDE HOLE, .103 TO .105 INCHES DIAMETER
10. ZERO READING INDICATOR

Figure 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-82684(OS) SPRING, HELICAL TORSION

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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